

Synthesis Fit Validation Study

Family-normalized code validation and raw pair-level human validation

Executive Summary

This report contains two separate validation studies. Code validation is a large-sample occupational-family proxy test. Human validation is a smaller blinded pair-level test against independent rubric judgments. They use the same structured weights and underlying scorers, but they use different aggregation, units, metrics, and claims.

Study	Question answered	Evaluation unit	Primary evidence
Code validation	Does a resume score highest against real JDs from its own occupational family?	353 resumes evaluated across 269 parsed JDs in three families.	Top-1 accuracy, mean rank, MRR, margin, and family confusion.
Human validation	Do arm scores rank specific resume-JD pairs in the same direction as human judgments?	54 unique real resume-JD pairs reviewed in one blinded session.	Spearman, Kendall tau-b, pairwise ordering, and diagnostic label metrics.

- **Code validation result:** the best pre-specified method is Family-normalized hybrid 0.25 rules / 0.75 semantic, with top-1 accuracy of **68.0%**, MRR of **0.825**, and mean rank of **1.41**.

- **Human validation result:** Hybrid 0.25 rules / 0.75 semantic produced Spearman **0.294** and pairwise ordering **0.609**, the highest values for those measures among the five pre-specified arms.

- **Combined decision:** the two studies support retaining hybrid 0.25. They do not establish that every individual fit score or product score band is accurate.

- **Change scope:** this report does not add a score transformation, new threshold, or production deployment.

Part I - Code Validation

Large-sample occupational-family proxy evaluation

Code Validation Question

The production Fit Analyzer is a one-JD-to-one-resume scorer. A family-level task cannot prove that a specific score is correct for a specific pair. Instead, this validation asks a narrower question: does a scoring method generally place a resume closer to postings from its own occupational family than to postings from other scoped families?

Code Validation Design

- Input data: local resume and posting datasets stored under the repo's gitignored Datasets directory.
- Family mapping: candidate postings are classified by the LLM mapper into 21 retained families plus UNMAPPED; the current validation filters to Consultant, Finance, and Information Technology.
- JD sampling: 300 high-confidence postings are collected across the three scoped families before parser gating.
- Parser gate: selected JDs are parsed with production parseJD(); the main study keeps JDs with at least three parsed requirements.
- Scoring unit: each resume is scored against every retained JD. Raw pair scores are averaged by JD family.
- Hybrid aggregation: the structured and semantic family-average maps are independently min-max normalized per resume, then blended. The highest resulting family score is the prediction.
- Embedding evidence: strict BGE backend, model Xenova/bge-small-en-v1.5, requested revision ea104dacec62c0de699686887e3f920caeb4f3e3 enforced for remote loading, packaged-model bundle SHA-256 3aacb9244b59b4b9..., with no fallback.

Code Validation Dataset After Parsing

Family	Resumes	JDs kept	JDs originally sampled
Consultant	115	95	100
Finance	118	94	100
IT	120	80	100
Total	353	269	300

The parseability gate kept **269** of **300** postings. The threshold was at least **3** parsed requirements per JD.

Family	Kept	Dropped	Mean req.	Median req.	Zero req.
Consultant	95/100	5	12.65	14.0	0
Finance	94/100	6	13.22	14.0	0
IT	80/100	20	8.96	7.5	3

Code Validation Methods Compared

Method	Description	Production interpretation
Rules-only structured	Current deterministic scoreFit() logic: O*NET-grounded skill extraction, requirement status, gaps, and evidence.	Strong baseline and most interpretable path.
Embedding-only semantic	Requirement-level semantic retrieval over resume evidence chunks using local BGE-small embeddings.	Tests whether semantic similarity aligns better with human-like fit judgement.
Hybrid arms	Per-resume family-normalized blend of structured and semantic family-average maps, using structured weights 0.25, 0.50, and 0.75.	A proxy aggregation for occupational-family discrimination; it is not the production pair formula.

Code Validation Headline Results

Arm	Top-1	Mean rank	MRR	Mean margin
Rules-only structured	54.7%	1.63	0.743	2.60
Embedding-only semantic	65.2%	1.43	0.813	0.27
Family-normalized hybrid 0.25 rules / 0.75 semantic	68.0%	1.41	0.825	0.18
Family-normalized hybrid 0.50 rules / 0.50 semantic	64.3%	1.47	0.803	0.18
Family-normalized hybrid 0.75 rules / 0.25 semantic	56.1%	1.58	0.756	0.12

Top-3 accuracy is intentionally not reported as a headline metric because this validation has only three families. The more informative metrics are top-1, mean rank, MRR, margin, and the confusion matrix.

Code Validation Per-Family Findings

Family	Structured	Embedding	Hybrid 0.25
Consultant	39.1%	34.8%	38.3%
Finance	100.0%	78.0%	89.0%
IT	25.0%	81.7%	75.8%

Finance is the strongest family, embedding materially improves IT, and Consultant is the hardest family because business, finance, and consulting language overlap.

Code Validation Confusion Matrix - Best Arm

Rows are true resume families; columns are predicted families. Values are resume counts.

True / Predicted	Consultant	Finance	IT
Consultant	44	21	50
Finance	3	105	10
IT	22	7	91

Code Validation Interpretation

- The family-normalized hybrid 0.25 arm is the best pre-specified family-level proxy method at 68.0% top-1 accuracy. It balances semantic flexibility with rules-based grounding.
- No formal success threshold was pre-registered for this proxy study, so the result is reported as a comparative arm outcome rather than a pass/fail claim.
- Consultant is the main failure mode because its language overlaps with the other scoped families. Some cross-family errors may still represent reasonable transferability in the production task.

Code Validation Limitations

- **Proxy task:** family top-1 measures coarse occupational discrimination, not one-pair fit accuracy.
- **Labels and scope:** posting families come from the LLM mapper, and only three families are included.
- **Parser and transferability:** parser errors can affect scores, while reasonable cross-family fits are counted as wrong by the proxy task.

Part II - Human Validation

Blinded pair-level comparison against rubric-based human judgments

Human Validation Question and Design

The pair-level study tests the production question more directly: whether a specific resume is a weak, medium, or strong fit for a specific real JD. The reviewer completed all judgments before arm scores or arm labels were merged into the comparison file.

- **Source:** the same frozen real-data pool used by code validation: 353 eligible resumes and 269 eligible JDs.
- **Candidate pool:** 5832 scored combinations; 54 unique resumes and 54 unique JDs were selected.
- **Stratification:** 6 pairs in every JD-family by LOW/MID/HIGH band cell (3 families x 3 bands x 6 pairs), including 18 cross-family stress pairs.
- **Selection priority:** within each cell, pairs with greater disagreement across the five arms were selected first. The sample is designed for method comparison, not prevalence estimation.
- **Blinding:** one review session; arm scores, arm labels, source-family fields, and selection bands were hidden during review. No second session was included.
- **Arms:** structured, embedding, hybrid 0.25, hybrid 0.50, and hybrid 0.75. Semantic scoring used strict local BGE-small embeddings with no fallback.
- **Pair formula:** each hybrid directly blends raw 0-100 structured and semantic pair scores. No min-max normalization, score transformation, or calibration is applied.

Human Validation Labelling Rubric

Dimension	0	1	2
Core requirements	Few core requirements met	Some met with meaningful gaps	Most important requirements met
Evidence quality	Little relevant evidence	Adjacent or weak evidence	Direct and specific evidence
Seniority and scope	Material mismatch	Partly comparable	Comparable level and complexity
Gating requirements	Clear gating item absent	Uncertain or partly met	No material gating gap

The four scores sum to a human total from 0 to 8. Totals 0-2 are Weak, totals 3-5 are Medium, and totals 6-8 are Strong. A zero on gating requirements caps the label at Medium. The comparison file contains only these four rubric scores, the total, the human label, and the five arm score/label pairs.

Human Validation Label Distribution

Label	Count	Share
Weak	26	48.1%
Medium	18	33.3%
Strong	10	18.5%
Total	54	100.0%

Human Validation Five-Arm Results

Arm	Spearman	Kendall	Pairwise	Balanced acc.	Macro F1	Ordinal MAE
Rules-only structured	0.223	0.171	0.590	0.333	0.217	0.704
Embedding-only semantic	0.276	0.220	0.606	0.333	0.104	1.296
Hybrid 0.25 rules / 0.75 semantic	0.294	0.213	0.609	0.337	0.234	0.815
Hybrid 0.50 rules / 0.50 semantic	0.247	0.183	0.595	0.444	0.381	0.537
Hybrid 0.75 rules / 0.25 semantic	0.218	0.159	0.582	0.370	0.289	0.648

Hybrid 0.25 rules / 0.75 semantic has the highest Spearman correlation (0.294) and pairwise ordering accuracy (0.609). Embedding-only semantic has the highest Kendall tau-b (0.220), compared with 0.213 for hybrid 0.25. The correlations are positive but modest, so the result supports relative arm selection rather than a claim of high absolute agreement.

Human Validation Interpretation

- **Pair-level rank evidence:** hybrid 0.25 is best on the two most decision-relevant human-study ranking measures: Spearman correlation and pairwise ordering accuracy.
- **Kendall qualification:** Embedding-only semantic has a slightly higher Kendall tau-b than hybrid 0.25, so no single arm leads every rank measure.
- **Threshold metrics do not overturn the decision:** Hybrid 0.50 rules / 0.50 semantic has the highest balanced accuracy under the pre-specified three-level validation mapping at 45 and 65. The mapping combines the product's 65-79 and 80+ presentation bands as Strong, so it does not validate the 80 cut point separately. These metrics remain secondary because they are highly dependent on each arm's raw score scale.
- **Human-study conclusion:** among the five pre-specified arms, the primary pair-level rank evidence supports hybrid 0.25, while the modest effect sizes limit claims of absolute agreement.
- **No new calibration:** human totals were not multiplied or transformed, and the product score boundaries were not changed or tuned after review.

Human Validation Limitations

- **Single reviewer:** the study measures agreement with one blinded judgment set and does not estimate inter-rater reliability.
- **Stratified sample:** equal LOW/MID/HIGH cells and within-cell disagreement enrichment support arm comparison but do not estimate natural production prevalence.
- **Modest association:** all rank correlations are below 0.30, so the study does not establish strong human-system agreement.
- **Scale effects:** the five arms occupy different raw score ranges, making fixed-boundary label metrics less suitable than rank measures for choosing among arms in this study.
- **Human labels:** written resume and JD evidence was reviewed without external credential verification.

Combined Decision and Reporting Position

How the two validation studies contribute to the final decision

Study	What it supports	What it does not establish
Code validation	Relative arm performance on coarse occupational-family discrimination at larger sample scale.	Correctness of a specific resume-JD score or the validity of product score bands.
Human validation	Relative arm performance on pair-level ordering against 54 blinded rubric judgments.	Natural production prevalence, inter-rater reliability, or absolute score-band validity.

Code validation provides the larger-sample evidence that hybrid 0.25 preserves the strongest coarse occupational-family signal among the pre-specified arms. Human validation provides the more task-direct evidence that hybrid 0.25 leads the main pair-level ranking measures. Together they support retaining hybrid 0.25 as the production Fit Analyzer arm. Neither study establishes that the current score bands are human-validated, and no new thresholds were introduced.

Reproducibility: raw study artifacts remain offline and gitignored. De-identified summary metrics, manifests, and checksums are stored under reports/fit-validation so the reported evidence can be audited without publishing resume or job-description text.